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Models for a series of studies based on geometrical representation

Aníbal Areia[a], Manuela M. Oliveira[b,*], João T. Mexia[c]

[a] *Department of Economy and Management, College of Business Administration, Polytechnic Institute of Setúbal, Portugal*
[b] *Department of Mathematics, University of Èvora, Colégio Luis António Verney, Rua Romão Ramalho 59, 7002 Èvora, Portugal*
[c] *Department of Mathematics, FCT, Nova University of Lisbon, Quinta da Torre, 2825 Monte da Caparica, Portugal*

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Abstract

The Escoufier operators are used to obtain a geometrical representation for the studies in a series. A model is developed assuming the normality of the vec of those operators. It is shown how to validate such a model. An application to the results of local elections in mainland Portugal is used for assessing the proposed model.
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1. Introduction

In the last few decades, many authors have studied a series of double-entry tables, thus originating the Joint Analysis of Tables (ACT).

We can consider the STATIS methodology as an ACT method that uses Euclidian distances between configurations observed in k situations. This approach was introduced by L'Hermier des Plantes [3] and developed by many authors, amongst them [2]. These methods use triplets of matrices $(\mathbf{X}_i, \mathbf{D}_{pi}, \mathbf{D}_n)$, which we designate as studies. Thus, a study consists of a data matrix $\mathbf{X}_i$ with a line for objects and a column for variables and of two weight matrices $\mathbf{D}_n$ and $\mathbf{D}_{mi}$ for objects and variables.

* Corresponding author. Tel.: +351 266 745300; fax: +351 266 744546.
E-mail addresses: aareia@esce.ips.pt (A. Areia), mmo@uevora.pt (M.M. Oliveira).

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Escoufier [1] defined the matrices $\mathbf{A}_i = \mathbf{X}_i\mathbf{D}_{p_i}\mathbf{X}_i^t\mathbf{D}_n, i = 1,\ldots,k$, and the matrices $\mathbf{B}_i = \mathbf{X}_i^t\mathbf{D}_{n_i}\mathbf{X}_i\mathbf{D}_p$, $i = 1,\ldots,k$ to represent a series of studies of the first and second types, respectively. The STATIS methodology defines $\mathbf{S} = \left[s_{ij}\right]_{\substack{i=1,\ldots,k \\ j=1,\ldots,k}}$ as the interstructure matrix with elements being the Hilbert–Schmidt scalar products $(\mathbf{A}_i \mid \mathbf{A}_j) = \mathrm{tr}(\mathbf{A}_i\mathbf{A}_j)$, $i, j = 1,\ldots,k$ [series of first type] and $(\mathbf{B}_i \mid \mathbf{B}_j) = \mathrm{tr}(\mathbf{B}_i\mathbf{B}_j)$, $i, j = 1,\ldots,k$ [series of second type]. The matrix $\mathbf{S}$ will be symmetric, with eigenvalues $\theta_1,\ldots,\theta_k$, and mutually orthogonal eigenvectors $\boldsymbol{\gamma}_1,\ldots,\boldsymbol{\gamma}_k$.

When the representation points are located along the first axis, the series of studies has a common structure (see [2]). Therefore the coordinates of those points with indexes $j = 2,\ldots,k$ will have small h and, since $\theta_j = \theta_j \left\|\boldsymbol{\gamma}_j\right\|$, $j = 1,\ldots,k$, the $\theta_2,\ldots,\theta_k$, will be small; thus

$$S = \sum_{j=1}^{k} \theta_j \boldsymbol{\gamma}_j \boldsymbol{\gamma}_j^t \approx \theta_1 \boldsymbol{\gamma}_1 \boldsymbol{\gamma}_1^t = \left(\theta_1^{1/2}\boldsymbol{\gamma}_1\right)\left(\theta_1^{1/2}\boldsymbol{\gamma}_1\right)^t,$$

with $\theta_1\boldsymbol{\gamma}_1\boldsymbol{\gamma}_1^t$ being a rank one matrix. More generally, if the representation points are near the space spanned by $\boldsymbol{\gamma}_1,\ldots,\boldsymbol{\gamma}_k$, with $h < k$, the series of studies will have a degree h common structure and $S \approx \sum_{j=1}^{h} \theta_j\boldsymbol{\gamma}_j\boldsymbol{\gamma}_j^t$ with $\sum_{j=1}^{h} \theta_j\boldsymbol{\gamma}_j\boldsymbol{\gamma}_j^t$ a rank h matrix. We are thus led to the model

$$\mathbf{S} = \sum_{j=1}^{h} \lambda_j \boldsymbol{\alpha}_j \boldsymbol{\alpha}_j^t + \mathbf{E}$$

where the $\boldsymbol{\alpha}_1,\ldots,\boldsymbol{\alpha}_k$ are pairwise orthogonal norm 1 vectors and $\mathbf{E}$ is an error matrix. We will indicate that $\mathbf{Z}$ is normal with the mean vector $\boldsymbol{\eta}$ and the variance-covariance matrix $\sigma^2\mathbf{W}$ by having $\mathbf{Z} \sim N(\boldsymbol{\eta}, \sigma^2\mathbf{W})$ and that $\mathbf{Z}$ is approximately normal with mean vector $\boldsymbol{\eta}$ and variance-covariance matrix $\sigma^2\mathbf{W}$, by having $\mathbf{Z} \overset{\circ}{\sim} N(\boldsymbol{\eta}, \sigma^2\mathbf{W})$. Oliveira and Mexia [6,7] considered the case $h = 1$ and assumed that $\mathrm{vec}(\mathbf{E}) \sim N(\mathbf{0}, \sigma^2\mathbf{I}_{k^2})$. This model, though manageable, did not account for the symmetry of $\mathbf{S}$. Oliveira and Mexia [9,10], considered that $\mathbf{E} = \mathbf{E}^0 + \mathbf{E}^{o\mathbf{t}}$, with $\mathrm{vec}(\mathbf{E}^0) \sim N(\mathbf{0}, \sigma^2\mathbf{I}_{k^2})$. However, the introduction of errors at the matrices' $\mathbf{S}$ level makes these formulations somewhat artificial; so, we now introduce errors at the level of matrices $\mathbf{A}_i, i = 1,\ldots,k$ in the series of the first type and of the matrices $\mathbf{B}_i, i = 1,\ldots,k$ in the series of the second type. Since the treatment for both types of series is the same we will consider only first type series. The model presented in this paper applies to a larger family of series than the former ones and rests on less strict assumptions. Theeir advantages are obtained at the cost of heavier developments. The model is developed while comparing it with the first symmetrical model in which we have $\mathbf{E} = \mathbf{E}^0 + \mathbf{E}^{o\mathbf{t}}$.

2. Model formulation

2.1. First symmetrical model

For comparison's sake, we now present briefly the first symmetrical model and its validation [6].

When the model holds, we have

$$\mathbf{S} = \lambda\boldsymbol{\alpha}_1\boldsymbol{\alpha}_1^t + \mathbf{E},$$

with $E = [e_{ij}]$ when $e_{ij} = e_{ji}$, and the

$$e_{11}, \ldots, e_{kk}\sqrt{2}e_{12}, \ldots, \sqrt{2}e_{1k}, \sqrt{2}e_{23}, \ldots, \sqrt{2}e_{(k-1)k},$$

are independent and identically distributed (i.i.d.) normal with null mean values and variance σ^2. Numerical studies showed that when $k \leq 20\tau = \frac{\lambda}{\sigma^2} > 200$, we can use the first eigenvalue θ_1 and the first eigenvector $\boldsymbol{\gamma}_1$ of $\mathbf{S}$ as estimators of λ and $\boldsymbol{\alpha}$, so that

$$\mathbf{W} = \mathbf{S} - \theta_1 \boldsymbol{\gamma}_1 \boldsymbol{\gamma}_1^t \approx \mathbf{S} - \lambda_1 \boldsymbol{\alpha}_1 \boldsymbol{\alpha}_1^t.$$

Thus, with $m = \frac{k(k+1)}{2}$, if the model holds, the random variables

$$Y_1 = w_{11}, \ldots, Y_k = w_{kk}, Y_{k+1} = \sqrt{2}w_{12}, \ldots, Y_m = \sqrt{2}w_{(k-1)k},$$

will be i.i.d. $\sim N(0, \sigma^2)$. Thus for validating the model, we may test

$$H_0 : Y_1, \ldots, Y_m \text{ i.i.d. } \sim N(0, \sigma^2)$$

using, for instance, the Shapiro–Wilk test. An alternative test will have as statistic

$$F = \frac{(m-1)\left(\sum_{i=1}^{m} Y_i\right)^2}{m\sum_{i=1}^{m} Y_i^2 - \left(\sum_{i=1}^{m} Y_i\right)^2},$$

which, when H_0 holds, has the central F distribution with 1 and $m-1$ degrees of freedom. When H_0 does not hold, the $Y_1, \ldots, Y_m$ will have non null mean values $\mu_1, \ldots, \mu_m$, and the numerator and denominator of F will have non centrality parameters $\delta_1 = m\left(\frac{\overline{\mu}}{\sigma}\right)^2$ and $\delta_2 = \frac{1}{\sigma^2}\sum_{i=1}^{m}(\mu_i - \overline{\mu})^2 = \frac{1}{\sigma^2}\left(\|\boldsymbol{\mu}\|^2 - m\overline{\mu}^2\right)$, with $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_m)$ and $\overline{\mu} = \frac{1}{m}\sum_{i=1}^{m}\mu_i$. Hence, there will be alternatives to H_0 in which δ_1 predominates over δ_2 (δ_2 predominates over δ_1), and in which F tends to take larger (smaller) values than when H_0 holds. A numerical study showed that, with $z_{p,r,s}$ as the quantile for probability p of the central F distribution with r and s degrees of freedom, the acceptance region $[z_{\frac{p}{2},1,m-1}, z_{1-\frac{p}{2},1,m-1}]$ leads to p level tests of power increasing rapidly with δ_1 and δ_2 (e.g. [4]).

To validate this model, we expect that H_0 is not rejected while $H_0^\circ : \lambda_1 = 0$ is rejected. We point out that H_0° may be rewritten as $H_0^\circ : \tau = 0$ and that its alternative is $H_1^\circ : \tau > 0$.

Given a $k \times k$ matrix $\mathbf{B} = [b_{ij}]$, $\mathbf{w}(\mathbf{B})$ will have components

$$b_{22}, \ldots, b_{kk}, \sqrt{2}b_{23}, \ldots, \sqrt{2}b_{2k}, \ldots, \sqrt{2}b_{(k-1),k}.$$

Then, if $\mathbf{P}^\circ$ is an orthogonal matrix whose first row vector is $\boldsymbol{\gamma}_1$, the statistic

$$F^\circ = \frac{(k-1)\theta_1^2}{2\|\mathbf{w}(\mathbf{P}^\circ \mathbf{S} \mathbf{P}^{\circ\mathbf{t}})\|^2}, \tag{2.1}$$

will have the central F distribution with k and $\frac{k(k-1)}{2}$ degrees of freedom when H_0° holds (e.g. [10]).

Moreover, under the alternatives to H_0°, F° will be approximately distributed as an F statistic with k and $\frac{k(k-1)}{2}$, degrees of freedom and non centrality parameter δ. When the model is fully

validated, we can condense the information contained in the series of studies into an adjusted structure vector $\tilde{\boldsymbol{\beta}} = \theta_1 \boldsymbol{\gamma}_1$ and a sum of squares of residuals $V = \|\mathbf{w}(\mathbf{P}^{\circ}\mathbf{S}\mathbf{P}^{\circ t})\|^2$.

2.2. Approximate normality

In the first symmetrical model, we assume that $\mathbf{Z} = \text{vec}(\mathbf{S})$ is normal. We now intend to obtain conditions for having $\mathbf{Z}$ approximately normal. Let us assume that

$$\mathbf{A}_i = \mathbf{A}_i^0 + \mathbf{E}_i, \quad i = 1, \ldots, k$$

with $\mathbf{A}_i^0 = [a_{ilh}]$ fixed and $\mathbf{E}_i = [e_{ilh}]$ with $\text{vec}(\mathbf{E}_i) \sim \mathbf{N}\left(\mathbf{0}, \sigma^2 \mathbf{I}_{n^2}\right)$, $i = 1, \ldots, k$, these vectors being independent from each other. Then

$$\begin{aligned} S_{ij} &= \text{tr}\left(\mathbf{A}_i \mathbf{A}_j^t\right) \\ &= \sum_{l=1}^{n}\sum_{h=1}^{n} (a_{ilh} + e_{ilh})\left(a_{jlh} + e_{jlh}\right) \\ &= \sum_{l=1}^{n}\sum_{h=1}^{n} (a_{ilh}a_{jlh} + a_{ilh}e_{jlh} + a_{jlh}e_{ilh} + e_{ilh}e_{jlh}) \\ &= S_{ij}^{\circ} + \sum_{l=1}^{n}\sum_{h=1}^{n} e_{ilh}e_{jlh}, \quad i = 1, \ldots, k,\ j = 1, \ldots, h \end{aligned}$$

where $S_{ij}^{\circ} = \sum_{l=1}^{n}\sum_{h=1}^{n}(a_{ilh}a_{jlh} + a_{ilh}e_{jlh} + a_{jlh}e_{ilh})$.

Now, with $S_{ij}^{\circ} = [s_{ij}^{\circ}]$, $\text{vec}(\mathbf{S}^{\circ})$ will be normally distributed. Thus $\mathbf{Z}$ will be approximately normally distributed whenever we can discard the error products $\sum_{l=1}^{n}\sum_{h=1}^{n} e_{ilh}e_{jlh}, i = 1, \ldots, k,\ j = 1, \ldots, k$. In the Appendix, we present the results of simulations that show that when the variation coefficients $\frac{\sigma_j}{\mu_j}$ of a pair of independent normal variables X_j, $j = 1, 2$, are less then $\frac{1}{15}$, the X_j^2, $j = 1, 2$ and X_1X_2 may be treated as normal. Thus we will assume the elements of matrices $\mathbf{A}_i, i = 1, \ldots, k$ to have variation coefficients less then $\frac{1}{15}$, and take $S_{ij} \approx S_{ij}^{\circ}, i = 1, \ldots, k, \quad j = 1, \ldots, k$.

Our assumption about the variation coefficients does not represent a strong restriction, since whenever standardized measurement techniques are used the variation coefficients tend to be quite below the proposed threshold of $\frac{1}{15}$.

With $K_{ij} = \text{tr}(\mathbf{A}_i^0\mathbf{A}_j^{0t}) = \sum_{l=1}^{n}\sum_{h=1}^{n} a_{ilh}a_{jlh}$, given the independence and homoscedasticity of the $\text{vec}(\mathbf{E}_i)$, $i = 1, \ldots, k$, we will have

$$\text{Var}\left(\sum_{l=1}^{n}\sum_{h=1}^{n} a_{ilh}e_{jlh} + a_{jlh}e_{ilh}\right) = \sigma^2 \sum_{l=1}^{n}\sum_{h=1}^{n}\left(a_{ilh}^2 + a_{jlh}^2\right) = \sigma^2\left(K_{ii} + K_{jj}\right); \quad i \neq j$$

and

$$\text{Var}\left(\sum_{l=1}^{n}\sum_{h=1}^{n} 2a_{ilh}e_{ilh}\right) = 4\sigma^2 \sum_{l=1}^{n}\sum_{h=1}^{n} a_{ilh}^2 = 4\sigma^2 K_{ii}, \quad i = 1, \ldots, n, \tag{2.2}$$

so

$$S_{ij} \overset{\circ}{\sim} N\left(K_{ij}; \sigma^2\left(K_{ii} + K_{jj}\right)\right), \quad i \neq j,$$

and

$$S_{ii} \overset{\circ}{\sim} N\left(K_{ii}; 4\sigma^2 K_{ii}\right).$$

Likewise, we get

$$\begin{aligned}
&\mathrm{Cov}\left(S_{ii}; S_{ij}\right) = \mathrm{Cov}\left(S_{ii}; S_{ji}\right) \approx 2\sigma^2 K_{ij}, \quad i \neq j \\
&\mathrm{Cov}\left(S_{ii}; S_{ju}\right) = 0, \quad i \neq j \neq u \\
&\mathrm{Cov}\left(S_{ij}; S_{ji}\right) = \mathrm{Var}(S_{ij}) \approx \sigma^2\left(K_{ii} + K_{jj}\right); \quad i \neq j, i \neq j \\
&\mathrm{Cov}\left(S_{ij}; S_{il}\right) = \mathrm{Cov}\left(S_{ij}, S_{li}\right) \approx \sigma^2\left(K_{jl}\right), \quad i \neq j, j \neq l \\
&\mathrm{Cov}\left(S_{ij}; S_{lu}\right) = 0, \quad i \neq j, j \neq l, l \neq u.
\end{aligned} \tag{2.3}$$

Moreover, with $\boldsymbol{\eta} = \mathrm{vec}([K_{ij}])$, we will have

$$\mathbf{Z} = \mathrm{vec}(\mathbf{S}) \overset{\circ}{\sim} \mathbf{N}(\boldsymbol{\eta}, \mathbf{L}),$$

with the elements of $\mathbf{L}$ given by the expressions (2.2) and (2.3).

2.3. Structure vectors and residues

For a common structure of level h, we have h structure vectors with $\boldsymbol{\beta}_i = \lambda_i \boldsymbol{\alpha}_i$, $i = 1, \ldots, h$, and

$$\mathbf{S} = \sum_{i=1}^{h} \lambda_i \boldsymbol{\alpha}_i \boldsymbol{\alpha}_i^t + \mathbf{E} = \sum_{i=1}^{h} \boldsymbol{\beta}_i \boldsymbol{\alpha}_i^t + \mathbf{E},$$

with $\mathbf{E}$ an error matrix.

The first symmetrical model was developed for series of level 1. Simulation studies show that $\widetilde{\boldsymbol{\beta}} = \theta \boldsymbol{\gamma}_1$ is a good estimator for $\tilde{\boldsymbol{\beta}}_1$, when $k \leq 20$ and $\tau = \frac{\theta_1^2}{\|S\|^2 - \theta_1^2} \geq 200$ (e.g. [8]).

For the proposed model, representing by $\otimes$ the Kronecker matrix product, we have

$$\tilde{\boldsymbol{\beta}}_j = \mathbf{S}\boldsymbol{\gamma}_j = \left(\mathbf{I}_k \otimes \boldsymbol{\gamma}_j^t\right)\mathbf{Z}, \quad j = 1, \ldots, h$$

while

$$E(\widetilde{\boldsymbol{\beta}}_j) = \left(\mathbf{I}_k \otimes \boldsymbol{\gamma}_j^t\right)\boldsymbol{\eta}, \quad j = 1, \ldots, h,$$

and

$$\not{\Sigma}(\widetilde{\boldsymbol{\beta}}_j) = \left(\mathbf{I}_k \otimes \boldsymbol{\gamma}_j^t\right)\mathbf{L}\left(\mathbf{I}_k \otimes \boldsymbol{\gamma}_j\right), \quad j = 1, \ldots, h.$$

If the common structure has level h, then the $\widetilde{\boldsymbol{\beta}}_1, \ldots, \widetilde{\boldsymbol{\beta}}_h$ are good estimators of $\boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_h$ and we will have

$$\mathbf{E} = \mathbf{S} - \sum_{i=1}^{h} \boldsymbol{\beta}_i \boldsymbol{\alpha}_i^t \approx \mathbf{S} - \sum_{i=1}^{h} \widetilde{\boldsymbol{\beta}}_i \boldsymbol{\gamma}_i^t = \mathbf{S} - \sum_{i=1}^{h} \left(\mathbf{I}_k \otimes \boldsymbol{\gamma}_i^t\right)\mathbf{Z}\gamma_i^t,$$

and then

$$\mathbf{R} = \mathrm{vec}(\mathbf{E}) \approx \mathbf{Z} - \mathrm{vec}\left(\sum_{i=1}^{h} \left(\mathbf{I}_k \otimes \boldsymbol{\gamma}_i^t\right)\mathbf{Z}\boldsymbol{\gamma}_i^t\right),$$

with $\operatorname{vec}\left(\sum_{i=1}^{h}\left(\mathbf{I}_k \otimes \boldsymbol{\gamma}_i^t\right)\mathbf{Z}\boldsymbol{\gamma}_i^t\right) = \left(\boldsymbol{\gamma}_i \otimes I_k \otimes \boldsymbol{\gamma}_i^t\right)\mathbf{Z}$. Thus

$$\begin{aligned}\mathbf{R} &= \left[\mathbf{I}_{k^2} - \sum_{i=1}^{h}\left(\boldsymbol{\gamma}_i \otimes \mathbf{I}_k \otimes \boldsymbol{\gamma}_i^t\right)\right]\mathbf{Z}\\ &= \left[\mathbf{I}_{k^2} - \mathbf{W}\right]\mathbf{Z},\end{aligned}$$

with $\mathbf{W} = \sum_{i=1}^{h}\left(\boldsymbol{\gamma}_i \otimes I_k \otimes \boldsymbol{\gamma}_i^t\right)$ and $\mathbb{V}(\mathbf{R}) = \left(I_{k^2} - \mathbf{W}\right)\mathbf{L}\left(I_{k^2} - \mathbf{W}\right)^t$.

While the vector $\mathbf{R}$ can be considered as a residual vector, $\widetilde{\boldsymbol{\beta}} = \left[\widetilde{\boldsymbol{\beta}}_1^t, \ldots, \widetilde{\boldsymbol{\beta}}_h^t\right]^t$ will be the adjusted global structure vector. These two vectors play a fundamental role in the inference. Since they are not independent we are led to get a homoscedastic vector of residues independent from $\widetilde{\boldsymbol{\beta}}$. We start by getting an $\mathbf{AR}$ vector independent of $\widetilde{\boldsymbol{\beta}} = \mathbf{BZ}$. Since $\mathbf{Z}$, is normal and we have

$$\tilde{\boldsymbol{\beta}} = \begin{bmatrix} I_k \otimes \boldsymbol{\gamma}_1^t \\ \cdots \\ I_k \otimes \boldsymbol{\gamma}_h^t, \end{bmatrix}\mathbf{Z}$$

$\mathbf{AR}$ is independent of $\tilde{\boldsymbol{\beta}}$ if $\mathbb{V}(\mathbf{AR}; \tilde{\boldsymbol{\beta}}) = 0$.

With $c = \operatorname{rank}((\mathbf{I}_{k^2} - \mathbf{W})\mathbf{L}\mathbf{B}^t)$, applying the Gram–Schmidt orthonormalization to the column vectors $\left(I_{k^2} - \mathbf{W}\right)\mathbf{L}\mathbf{B}^t$, we get mutually orthogonal norm one vectors $\mathbf{z}_1, \ldots, \mathbf{z}_c$.

Let the $\boldsymbol{\delta}_i$ be a vector with k^2 components all null, but with the component with index i which is 1, $i = 1, \ldots, k^2$. Applying now the Gram–Schmidt orthonormalization process to the vectors $\mathbf{z}_1, \ldots, \mathbf{z}_c, \boldsymbol{\delta}_1 \ldots, \boldsymbol{\delta}_{k^2-c}$, we get, besides the $\mathbf{z}_1, \ldots, \mathbf{z}_c$, the row vectors $\mathbf{a}_{1,}, \ldots, \mathbf{a}_{k^2-c}$ of matrix $\mathbf{A}$. We will then have the covariance matrix

$$\mathbb{V}(\mathbf{AR}) = \mathbf{A}\left(I_{k^2} - \mathbf{W}\right)\mathbf{L}\left(I_{k^2} - \mathbf{W}\right)^t\mathbf{A}^t,$$

with non null eigenvalues $\zeta_1, \ldots, \zeta_g$, to which are associated the eigenvectors $\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_g$. Since the covariance matrices do not have negative eigenvalues, the matrix

$$\mathbf{G} = \mathbf{diag}\left(\zeta_1^{-1/2}, \ldots, \zeta_g^{-1/2}\right)[\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_g]^t$$

will be defined. It is easily seen that $\mathbb{V}(\mathbf{GAR}) = \mathbf{I}_g$, so $\mathbf{Y} = \mathbf{GAR}$ will be the homoscedastic residues vector independent of $\tilde{\boldsymbol{\beta}}$, $(V_h = \|\mathbf{Y}\|)$.

2.4. Model validation

We now renew the comparison between models. The hypothesis H_0 considered in Section 2.1 is now replaced by

$$H_{0h} : Y_1, \ldots, Y_g \text{ i.i.d.} \sim N\left(0, \sigma^2\right).$$

with $Y_1, \ldots, Y_g$ as the components of $\mathbf{Y}$. This is the same test as we considered for H_0 in Section 2.1; the sole difference is that m is replaced by g. Thus the $1 - p$ level test will have an acceptance region $[z_{\frac{p}{2},1,g-1}, z_{1-\frac{p}{2},1,g-1}]$.

We now point out that when H_{0h} is not rejected, we are led to accept that the degree of the structure does not exceed h. But then we could have a common structure of lower degree. We are thus led to test the significance of $\widetilde{\boldsymbol{\beta}}_h$.

For $\mathbf{M}^+$, the Moore–Penrose inverse of $\mathbf{M}$,, the quadratic form

$$\mathbf{Q}_h = \tilde{\boldsymbol{\beta}}_h[\left(I_k \otimes \boldsymbol{\gamma}_h^t\right)\mathbf{L}\left(I_k \otimes \boldsymbol{\gamma}_h\right)]^+\tilde{\boldsymbol{\beta}}_h,$$

is distributed as the product of $\boldsymbol{\sigma}^2$ by a chi-square with

$$\mathbf{c}_h = \operatorname{rank}\left[\left(I_k \otimes \boldsymbol{\gamma}_h^t\right)\mathbf{L}\left(I_k \otimes \boldsymbol{\gamma}_h\right)\right]^+,$$

degrees of freedom. This chi-square will be central when $\boldsymbol{\beta}_h^\circ = \mathbf{E}\left(\widetilde{\boldsymbol{\beta}}_h\right) = 0$ so that,

$$F_h^\circ = \frac{\mathbf{g}}{\mathbf{c}_h}\frac{\mathbf{Q}_h}{\|\mathbf{Y}\|^2},$$

will have a central F distribution with $\mathbf{c}_h$ and g degrees of freedom.

When $\boldsymbol{\beta}_h^\circ = \mathbf{E}\left(\widetilde{\boldsymbol{\beta}}_h\right) \neq 0$, both Q_h and F_h° will have a non-centrality parameter

$$\boldsymbol{\delta}_h^0 = \frac{1}{\sigma^2}\boldsymbol{\beta}_h^{0t}\left[\left(I_k \otimes \boldsymbol{\gamma}_h^t\right)\mathbf{L}\left(I_k \otimes \boldsymbol{\gamma}_h\right)\right]^+\boldsymbol{\beta}_h^0,$$

with $\boldsymbol{\beta}_h^0 = \mathbf{E}\left(\boldsymbol{\beta}_h\right)$.

Thus, the test with statistics F_h° for $H_{0h}^0 : \boldsymbol{\beta}_h^0 = 0$ will be unbiased. It is convenient to start by testing H_{0h}, since under the alternatives H_{1h}, $\|\mathbf{Y}\|^2$ will be the product by σ^2 of a chi-square with g degrees of freedom and with non-centrality parameter $\boldsymbol{\delta}_1 + \boldsymbol{\delta}_2$, and there will be a loss of power for the test with statistic F_h, (e.g. [5]). In applying these results, we must test for growing values of the degrees of common structure until we do not reject H_{0h} and reject H_{0h}°.

We point out that to test H_{01}° is equivalent to testing hypothesis H_{0h}°, since testing either hypothesis is equivalent to testing the existence of a deterministic component in the model.

Moreover, when

$$\tau = \frac{\theta_1^2}{\sum_{j=2}^{k}\theta_j^2}$$

is sufficiently large, the representation points are located along the first axis and so we may adjust the new model with $h = 1$. This will be the case for the series of studies we consider in the application. Thus we will be in the situation in which we have two level 1 models for each series, which renders it easy to carry out model comparison.

3. Application to local elections in Portugal

3.1. Presenting the problem

Our study is based on the results of the local elections carried out in Portugal, in the years of 1985, 1989, 1993, 1997, and 2001. Portugal's territory is divided into districts. These are divided into townships. Each township is further divided into parishes. For each election in each township, we had a data (objects x variables) array. The objects were the civil parishes and the variables were the main parties participating in the electoral process: the Social Democratic Party (PSD); Portuguese Socialist Party (PS); Central Social Democratic/Popular Party (CDS/PP); United Democratic Coalition where the Communist Party has a predominant role (CDU); Other parties (OUT); Blank and invalid ballots (B/N); Abstentions (ABS).

Table 1
Two first eigenvalues and the preponderance values

	V.Cas	P. Bar	Bra	A. Fé	Ave	S.M.F	Gua	Pin
θ_1	86.888	126.264	145.213	105.563	120.169	102.488	144.113	111.733
θ_2	0.719	0.672	0.485	1.248	0.597	0.533	0.972	0.792
τ	11401.4	29599.8	54427.3	7039.4	36052.9	24746.5	19042.7	14960.0
	Lei	Alc.	Por	Elv	Far	Sil	Bej	Mer
θ_1	115.010	124.364	104.169	144.970	141.857	97.299	129.321	143.055
θ_2	0.179	0.593	0.129	0.514	0.219	0.199	0.164	0.109
τ	131484.1	33739.7	98166.1	62641.7	121701.2	73105.6	392064.8	280434.8

(Viana do Castelo (V.Cas), Ponta da Barca (P.Bar), Bragança (Bra), Alfandega da Fé (A.Fé), Aveiro (Ave), Santa Maria da Feira (S.M.F.), Guarda (Gua), Pinhel (Pin), Leiria (Lei), Alcobaça (Alc), Portalegre (Por), Elvas (Elv), Faro (Far), Silves (Sil), Beja (Bej) and Mertola (Mer)).

In this study, four districts stretching from North to South along the coast and four others along the border with Spain were selected. For each of these districts, two townships were selected. One of them contained the district capital city. Thus we had good coverage of the territory.

We now apply both models to the 16 series of studies. The data array was D_n- centered. We considered identity weight matrices $\mathbf{D}_{n_i} = \mathbf{I}_{n_i}$ for parishes (objects) and $\mathbf{D}_{p_i} = \mathbf{I}_{p_i}$ for parties (variables).

The first eigenvalue is very highly predominant ($\tau > 5000$) for all series, so that we could consider for both models a degree 1 common structure for all series. This fact is important, since we had only one adjusted structure vector for every series, which is the same for both models. Thus the model comparison will rest on the squares of residuals since, theoretically, the new model rests on less artificial assumptions.

3.2. Models' adjustment and comparison

The estimated preponderance of the first eigenvalues are extremely high (Table 1).

Thus, we may adjust the series with degree 1 common structure for both models. Moreover these values are clearly above the threshold of 200, so we can accept the adjusted structure vectors presented in Table 2.

In Table 3, we present the values of the test statistics for validating both models and the sums of squares of residuals also for both models.

While both models are validated for all series, the sum of squares is always lesser for the new model.

4. Final remarks

We presented in this article a new approach for the modelling of the inter-structure of a series of studies in the ACT method. We compared this approach with a previous one. Besides its more realistic theoretical background, the new model led to more precise results when applied to 16 series of studies. In deriving the new model, we used the fact that given independent normal variables with low variation coefficients, their squares and cross products may be treated as normal (as shown in simulations presented in the appendix). In future work we intend to:

(1) develop inferences for matched series of studies, namely using ANOVA-like analyses (e.g. [9]).

Table 2
Adjusted structure vectors

Election year	V.Cas	P. Bar	Bra	A. Fé	Ave	S.M.F	Gua	Pin
1985	35.722	51.256	61.229	48.695	53.941	40.077	57.716	48.432
1989	37.095	56.419	63.300	40.778	56.815	43.042	66.348	41.315
1993	33.817	62.272	65.362	52.640	53.900	46.369	62.877	40.435
1997	42.156	56.821	65.989	46.876	49.211	49.900	71.605	59.374
2001	44.461	55.007	68.587	46.273	54.554	49.039	62.885	57.196
Election year	Lei	Alc.	Por	Elv	Far	Sil	Bej	Mer
1985	50.119	72.298	41.815	51.721	57.963	41.714	57.182	65.941
1989	52.948	50.326	49.264	49.985	67.139	45.439	55.938	66.916
1993	48.262	52.563	47.862	47.763	56.466	42.323	58.677	61.514
1997	51.991	48.495	46.030	76.315	68.640	39.776	58.982	63.639
2001	53.662	50.912	47.604	87.963	66.001	47.844	58.338	61.683

Table 3
F tests and sums of squares for the first symmetrical model and the new model

	First symmetrical model							
	V.Cas	P. Bar	Bra	A. Fé	Ave	S.M.F	Gua	Pin
F	0.175	0.137	0.250	0.131	0.121	0.194	0.199	0.250
F°	26 115	66 402	125 975	13 562	78 493	63 959	37 615	30 740
V	0.575	0.538	0.387	1.583	0.401	0.424	1.090	0.834
	Lei	Alc	Por	Elv	Far	Sil	Bej	Mer
F	0.217	0.259	0.233	0.310	0.244	0.261	0.231	0.173
F°	481 626	58 548	864 161	101 110	632 799	290 377	813 337	3065 925
V	0.101	0.458	0.111	0.336	0.165	0.129	0.085	0.073
	New model							
	V.Cas	P. Bar	Bra	A. Fé	Ave	S.M.F	Gua	Pin
F	2.864	4.250	0.477	0.583	0.038	0.003	0.060	0.046
F_1°	160 901	285 115	289 266	130 612	324 730	241 668	179 691	148 563
V_1	0.270	0.221	0.251	0.404	0.185	0.212	0.401	0.376
	Lei	Alc	Por	Elv	Far	Sil	Bej	Mer
F	0.258	0.668	2.056	3.985	1.595	1.138	0.069	1.238
F_1°	611 702	221 258	801 233	276 642	779 280	446 696	762 419	1740 4
V_1	0.094	0.281	0.065	0.262	0.084	0.110	0.043	0.005

(2) continue to study the almost normality for low degree polynomials of independent normal variables with low variation coefficients.

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Table A.1
Mean values for correlation coefficients

Variables	VC_2	$VC_1 = \frac{1}{5}$	$VC_1 = \frac{1}{15}$	$VC_1 = \frac{1}{30}$
X_1^2; $\mu_1^2 + 2\mu_1 Y_1$		0.996	0.999	0.999
X_1X_2; $\mu_1\mu_2 + \mu_1 Y_2 + \mu_2 Y_1$				
	$\frac{1}{5}$	0.990	0.998	0.999
	$\frac{1}{15}$	0.998	0.999	0.999
	$\frac{1}{30}$	0.999	0.999	0.999

Appendix

In this appendix, we present simulations that show that when the variation coefficients $VC_j = \frac{\sigma_j}{\mu_j}$ of a pair of independent normal variables X_j, $j = 1, 2$ are sufficiently small, we can treat the variables X_1^2 and X_1X_2 as normal. In this way, we justify the use of approximate normality made in Section 2.2. First we point out that putting $X_j = \mu_j + e_j$, $j = 1, 2$, we have

$$X_1^2 = \mu_1^2 + 2\mu_1 e_1 + e_1^2$$

and

$$X_1X_2 = \mu_1\mu_2 + \mu_1 e_2 + \mu_2 e_1 + e_1e_2.$$

It is easy to see that

$$\frac{\mu_1^2 + 2\mu_1 e_1 + e_1^2}{\mu_1^2 + 2\mu_1 e_1} \xrightarrow[|\mu|\to\infty]{p} 1$$

and that

$$\frac{\mu_1\mu_2 + \mu_1 e_2 + \mu_2 e_1 + e_1e_2}{\mu_1\mu_2 + \mu_1 e_2 + \mu_2 e_1} \xrightarrow[|\mu_1|;\ |\mu_2|\to\infty]{p} 1,$$

where $\xrightarrow{p} 1$ indicates stochastic convergence. Since the variables in the denominators are normally distributed, we are led to assume that, for sufficiently large $|\mu_1|$ and $|\mu_2|$, the variables in the numerators are also normally distributed.

Nevertheless, these results only refer to limit distributions and, for practical purposes, we need a threshold. To obtain such a threshold we carried out simulations. Since the threshold we are looking for refers to the variation coefficient, we will consider pairs (Y_1, Y_2) of independent standardized normal variables taking $X_j = Y_j + \mu_j$, $j = 1, 2$, so that the variation coefficients of the X_j, $j = 1, 2$, will be $VC_j = \frac{1}{\mu_j}$, $j = 1, 2$. First, we carried out 10 000 runs using Mathematica, 5.0, to obtain mean values for the correlation coefficients between X_1^2 and $\mu_1^2+2\mu_1 Y_1$ and between X_1X_2 and $\mu_1\mu_2+\mu_1 Y_2+\mu_2 Y_1$. These results are presented in Table A.1.

Thus when the variation coefficients fall below $\frac{1}{15}$, there is a very strict linear relation between X_1^2 and $\mu_1^2 + 2\mu_1 Y_1$ and between X_1X_2 and $\mu_1\mu_2 + \mu_1 Y_2 + \mu_2 Y_1$. Next, we also carried 10 000 runs to obtain histograms for X_1^2 and X_1X_2 for the same values of the variation coefficients Fig. A.1.

Thus when the variation coefficients fall below $\frac{1}{15}$, histograms for X_1^2 and X_1X_2 are very approximately normal. Last, we used a third batch of 10,000 runs to test for normality. In

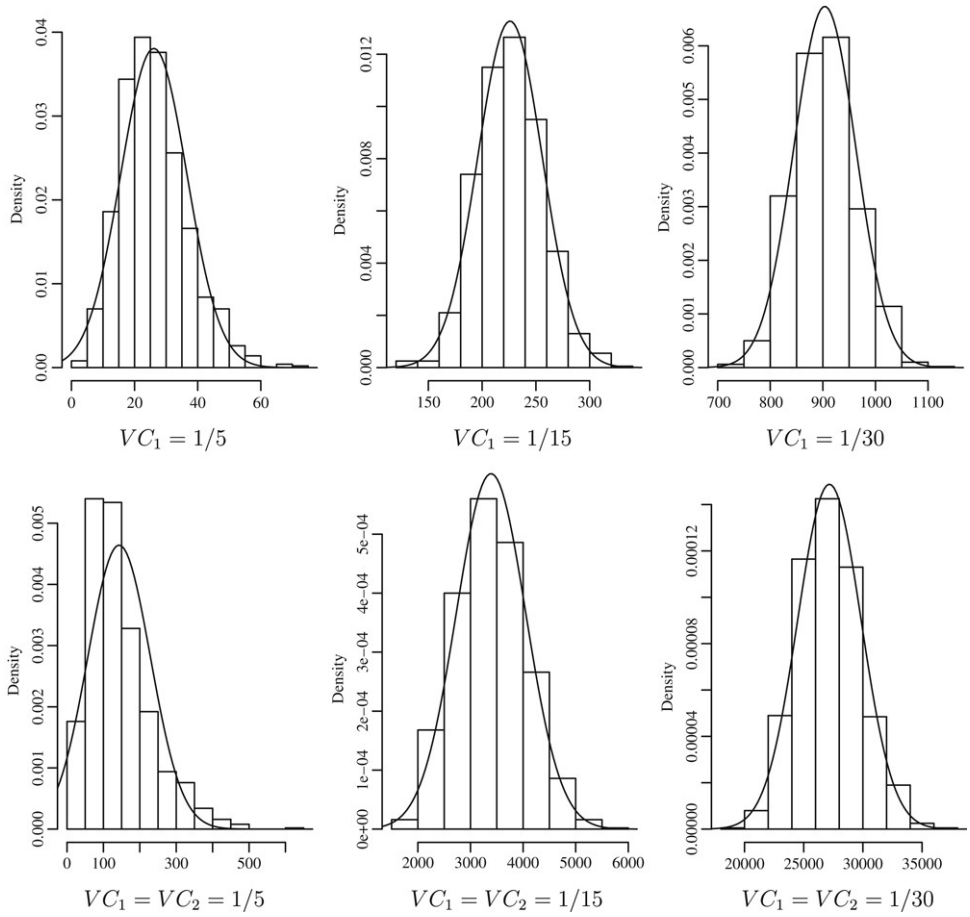


Fig. A.1. Histograms for variables X_1^2 and X_1X_2 for the values of the variation coefficients $\frac{1}{5}$, $\frac{1}{15}$, and $\frac{1}{30}$.

Table A.2
P-values of Kolmogorov–Smirnov normality test

Variables	$VC_1 = VC_2 = \frac{1}{5}$	$VC_1 = VC_2 = \frac{1}{15}$	$VC_1 = VC_2 = \frac{1}{30}$
X_1^2	0.033	0.529	0.720
X_1X_2	0.157	0.642	0.750

Table A.2 we have the corresponding average p-values for the Kolmogorov–Smirnov normality tests for X_1^2 and X_1X_2. Thus once the variation coefficients fall below the threshold of $\frac{1}{15}$ the normality of X_1^2 and X_1X_2 may be assumed.

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